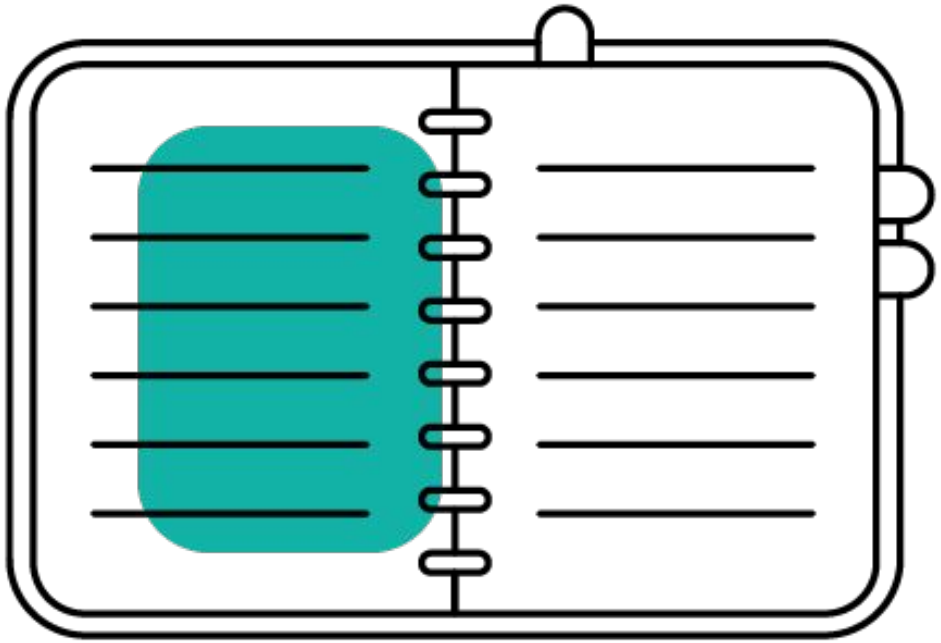
A Level
Design and Technology
Exam Support



Agenda

- Discuss the full range of questions that appear on the paper.
- Highlight answers at a range of levels, and identify key differences in reply.
- Identify how the mark scheme is applied.
- Discuss key tips for how to approach each type of question.



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Question verbs/command words

- Give/State
- Outline
- Explain (2 or 3 marks)
- Calculate
- Draw
- Describe using labelled sketches...
- Discuss
- Evaluate



Explain Questions – 2 marks

(b) The wardrobe is made from chipboard. A thin veneer of hardwood has been applied to the surface of the chipboard.

Explain **two** characteristics of veneered chipboard that make it a suitable material for flat pack wardrobes.

(4) Q01b 4

1 the veneer has an attractive grain which aids visuality and aesthetic appeal of the wardrobe, meaning sales may also increase due to aesthetics of the veneered surface.

2 Chipboard is dimensionally strong/stable and hence will remain robust for a long period of time and many use cycles of the wardrobe, this is important for when storing valuable items inside

MS bp.5

MS bp.2

Question number	Answer	Mark
1(b)	Any two explanations that include a correct characteristic and a linked justification of why the characteristic makes it a suitable material to use. 1. Available in large / wide / flat sheets (1) so no need to join materials / easily stored/transported in flat pack form (1) 2. Uniform strength / rigidity (1) so doors and side panels will not need supporting framework / will be able to withstand applied loads. (1) 3. Cheaper than solid wood / other manufactured boards (1) so can be used in budget products (1) 4. Considered sustainable (1) as it utilises offcuts and waste timber in its manufacture (1) 5. Gives the aesthetic appearance of a solid hardwood (1) without the associated cost / sustainability concerns / by hiding the base chipboard layer / attracts customers (1) 6. Stable material / uniform material (1) does not bend / warp / twist in a dry indoor environment (1) 7. Available in a range of different thicknesses (1) to suit the structural / stiffness requirements of the wardrobes (1) 8. Different veneers can be applied / used (1) to match / coordinate / contrast with other furniture (1)	(4)

Give a characteristic of chipboard and justify that one, therefore do not give several characteristics to gain the 2nd mark for that point.

Explain Questions – 3 marks

MS bp.3

(c) The chisel has a polypropylene handle.

Explain **one** benefit of using polypropylene for the chisel handle.

(3) Q01c 3

can be formed into a variety of shapes with a more ergonomic design so its more comfortable for the user. This also mean it can be used for longer without making the user uncomfortable.

Question number	Answer Additional guidance	Mark
1(c)	Any explanation that includes the identification of any benefit (1) and linked justifications of why that is a benefit (1+1): 1. Polypropylene is a tough material (1) it will not break/shatter/fracture (1) so has good impact resistance / providing durability / safety in use (1) 2. Polypropylene has a soft feel / is a warm material (1) improving the comfort / ease of use (1) ensuring a firm grip / good tool control (1) 3. Polypropylene can be moulded into complex shapes (1) allowing an ergonomic design to be easily produced / with grips (1) allowing the tool to be used, with comfort, for extended periods of time / reducing the risk of injury (1) 4. Polypropylene will not split or break with extensive use (1) because it is a durable material (1) due to its fatigue resistance (1) 5. Polypropylene can be coloured (1) making the tool more visible (1) allowing colour coding for identification (1) 6. Polypropylene is shock absorbent (1) which is effective when the chisel is hit with a hammer/mallet (1) reducing the risk of injury to the user / fatigue (1) 7. Polypropylene can be moulded onto the blade (1) so is unlikely to come loose in use (1) improving safety / ensuring longevity (1)	(3)

- Give a benefit and justify that one, therefore do not give several benefits in your answer.
- Ensure that the explanation/justification is more detailed and includes two relevant points.

(c) The chisel has a polypropylene handle.

Explain **one** benefit of using polypropylene for the chisel handle.

(3) Q01c 2

Polypropylene is impact resistant meaning when the chisel goes into the hammer it will not shatter and will be able to withstand the dynamic shock it experiences.

MS bp.6

(c) Explain **two** reasons why mahogany might **not** be considered a sustainable material.

(cacus) **(6)** Q03c 6

1 mahogany grows in tropical conditions which are very far away from many countries. This means that there will be more transport which will be further and will cause more fossil fuels to be emited in transport which isn't good for enviroment.

2 The woodland from mahogany cones from often isn't managed, this means that it is susceptible to overlogging and deforestation. It also means that once plant has been taken it may not be replanted

(Total for Question 3 = 12 marks) Total 12

MS bp.1

MS bp.2

(c) Explain **two** reasons why mahogany might **not** be considered a sustainable material.

(cacus) **(6)** Q03c 6

1 mahogany grows in tropical conditions which are very far away from many countries. This means that there will be more transport which will be further and will cause more fossil fuels to be emited in transport which isn't good for enviroment.

2 The woodland from mahogany cones from often isn't managed, this means that it is susceptible to overlogging and deforestation. It also means that once plant has been taken it may not be replanted

(Total for Question 3 = 12 marks) Total 12

MS bp.1

MS bp.2

Question number	Answer	Additional guidance	Mark
3(c)	Any two explanations that include identification of a reason (1) and linked justifications of that reason (1) + (1). 1. Mahogany grows in tropical regions / equatorial rain forests (1) meaning it is transported large distances to market (1) thus increasing carbon emissions / impacting on the carbon footprint (1) 2. Logging of mahogany has led to deforestation (1) causing loss of animal habitats / damaging ecosystems (1) and leaving barren land that is not replanted / reducing oxygen regeneration/CO_2 absorption (1) 3. Despite being protected by law / ethics (1) the value of the timber is so great (1) that illegal logging occurs (1) 4. Mahogany is a hardwood / comes from a deciduous tree (1) so is slower growing than softwoods (1) therefore cannot be considered as an easily renewable resource / is a finite material (1) 5. When produced commercially as a monocropping operation (1) the acid soil associated with the growth of mahogany (1) doesn't support ecosystems so the forests are almost devoid of	Do not accept repeated justification.	(6)

Draw Questions – Isometric/Perspective/Ortho Projection

7 Figure 6 shows an isometric projection of a component.

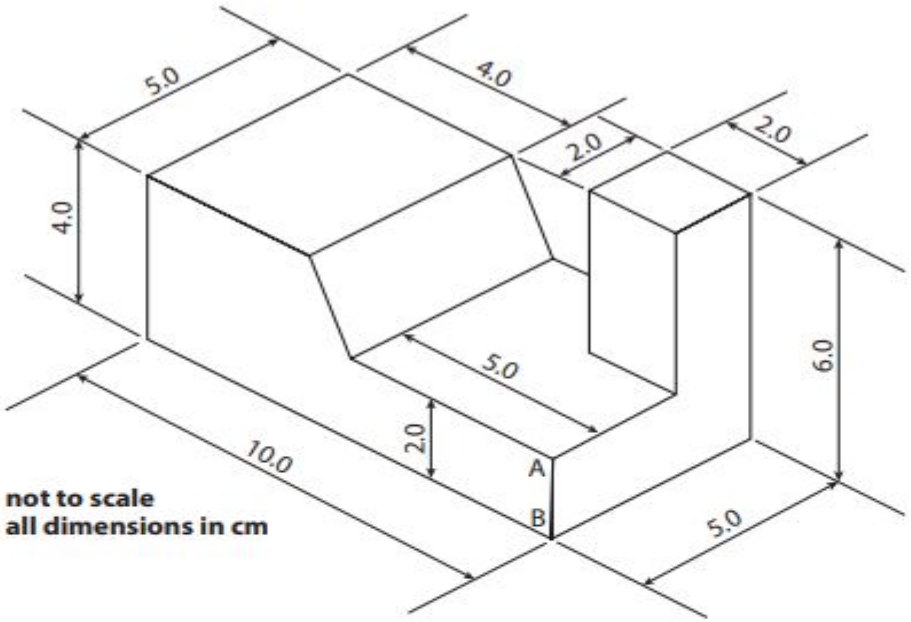


Figure 6

Isometric Projection

Designers use a range of different drawing techniques to convey their design ideas.

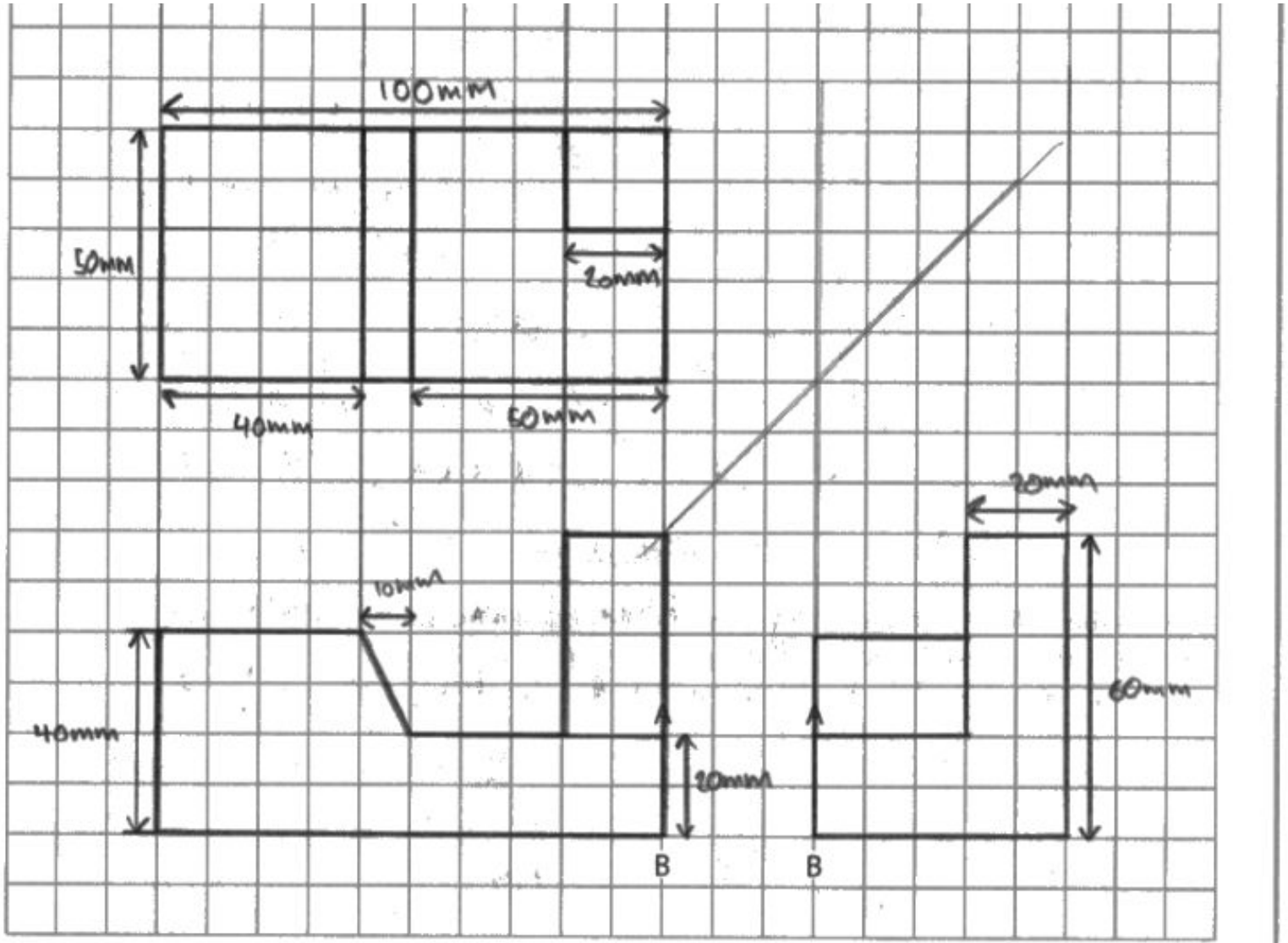
Draw an accurate 3rd angle orthographic projection of the component.

Use the grid provided at a scale of 1 square = 1 cm.

You should start at line A-B which has been shown for both the front and side elevation.

You must include sufficient dimensions on your drawing to allow a 3rd party to produce the component.

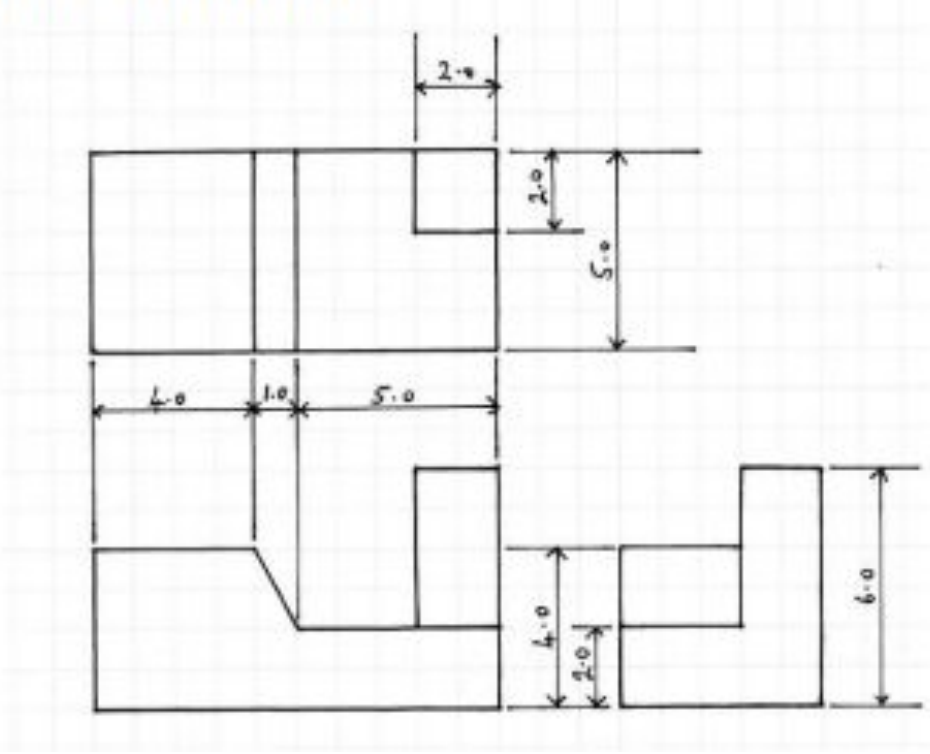
(6)

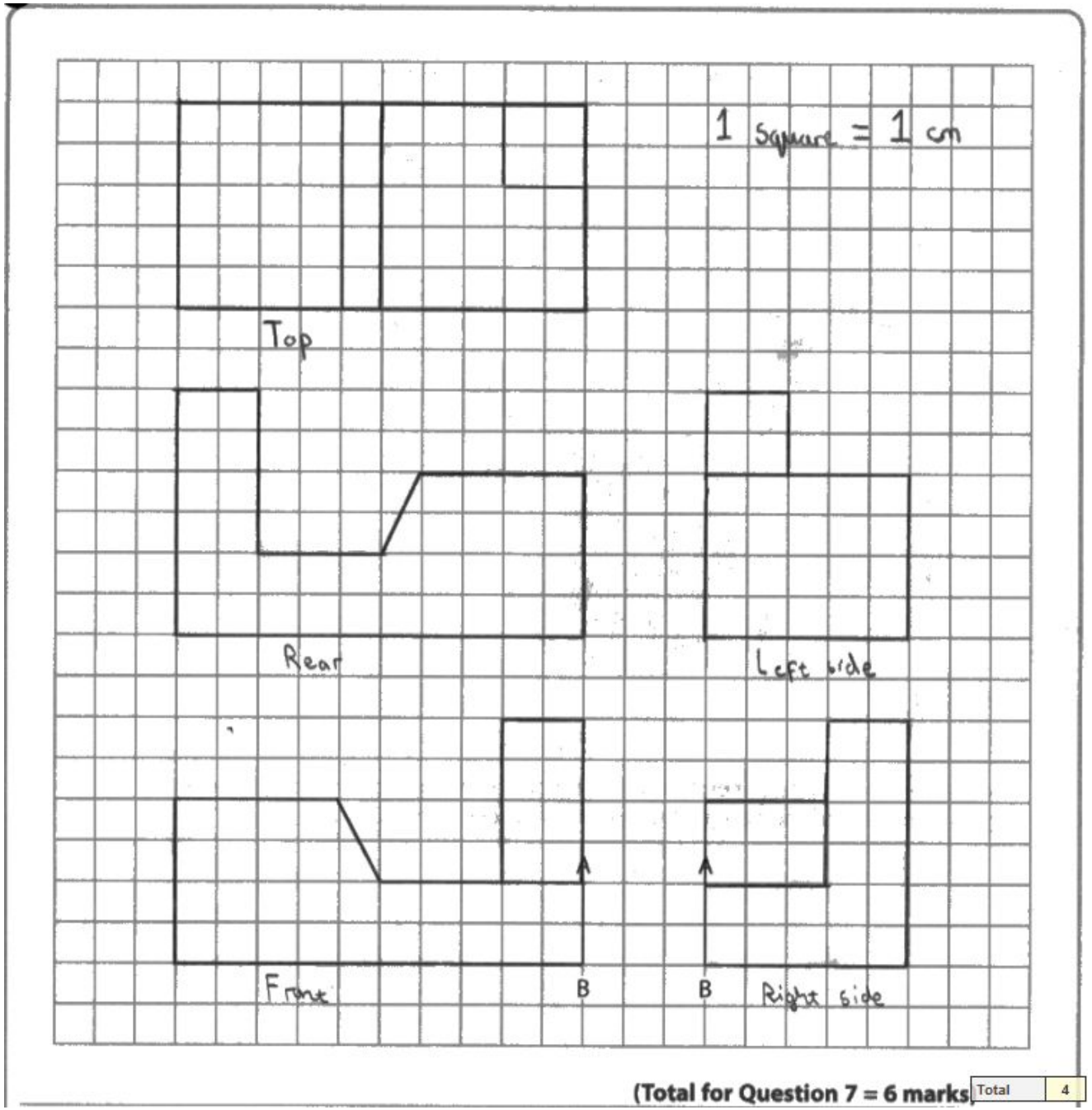


(Total for Question 7 = 6 marks)

Total	6

		inappropriate throughout.
Level 2	3 – 4	• Drawing is produced with some precision and accuracy. • Mostly appropriate use of orthographic projection techniques with mostly accurate orientation, layout and views • Some attention to scale, size, alignment and dimensions • Line style is mostly appropriate and consistent throughout.
Level 3	5 - 6	• Drawing is produced with precision and accuracy. • Fully appropriate use of orthographic projection techniques with correct orientation, layout and views • Accurately drawn to scale with correct dimensioning and alignment • Line style is appropriate and consistent throughout enhancing the 3D effect.

Question number	Answer	Mark
7	A third angle orthographic drawing of the component. The dimensions shown in the exemplar below provide the minimum number of discrete dimensions required to manufacture the product.  	(6)

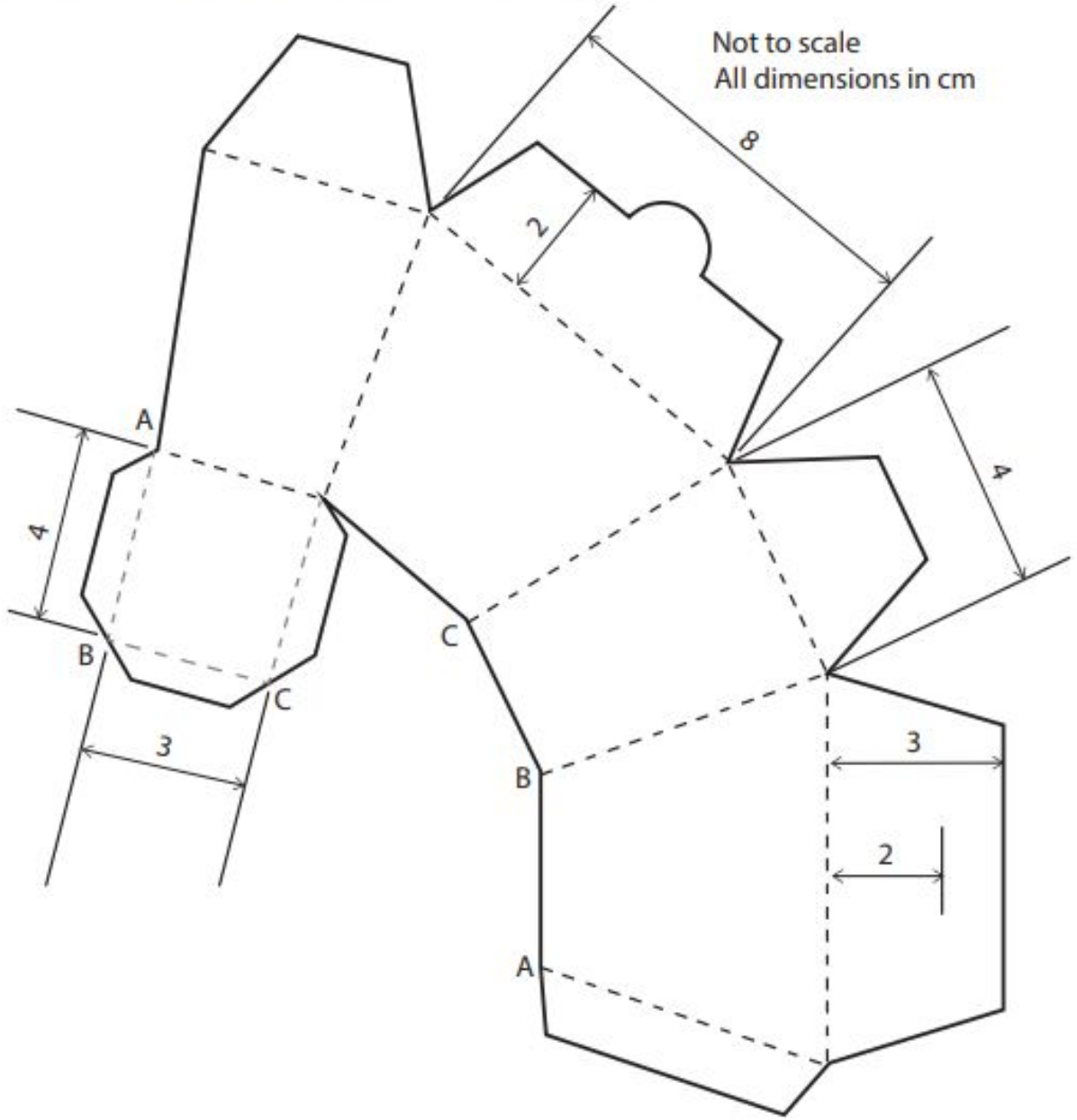


(Total for Question 7 = 6 marks)

Total	4

- Ensure that the students drawing/lines are clear and bold enough to be seen easily when scanned.
- Use the information/dimensions that are included in the question

7 Figure 6 shows the net (development) for a chocolate box.



Draw an accurate isometric projection of the chocolate box.

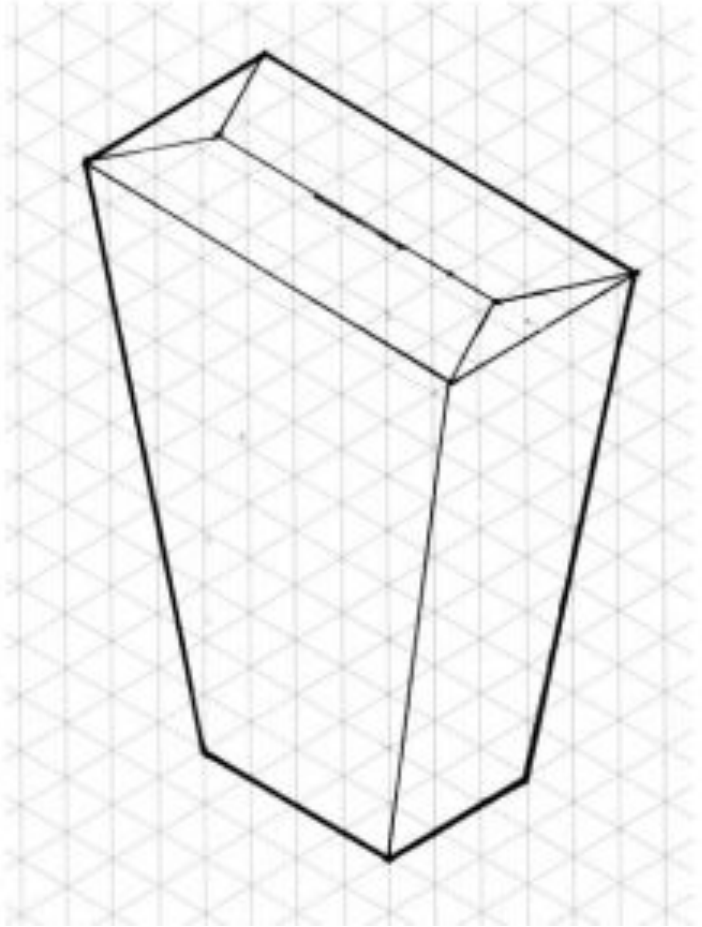
Use the isometric grid provided Starting with Lines A–B and B–C as your baselines.

Each segment of the grid equals 1 cm.

The height of the assembled box is 10 cm.

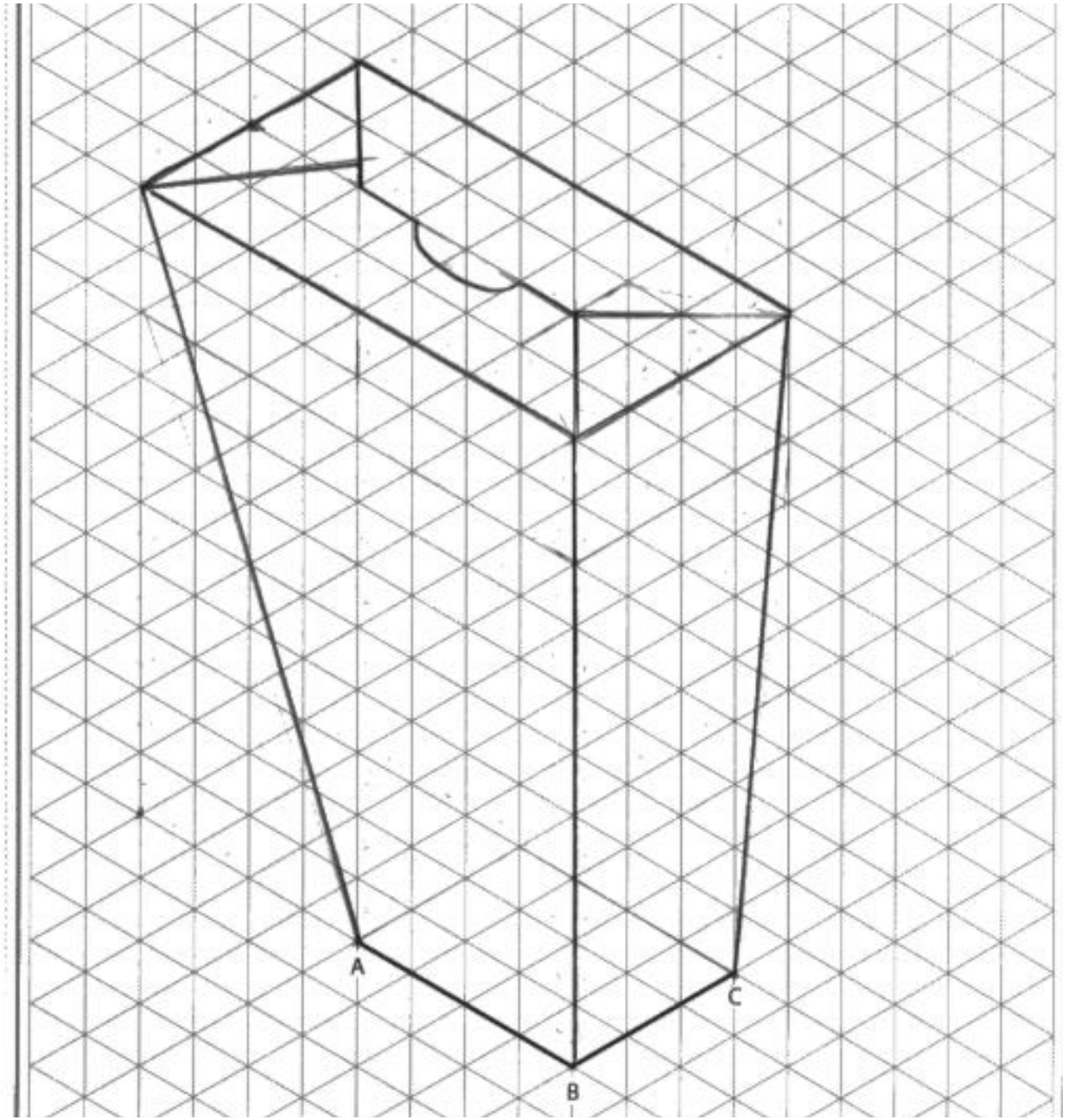
Dimensioning of the isometric projection is not required.

(6)

Question number	Answer	Mark
7	An isometric drawing of the chocolate box. 	(6)

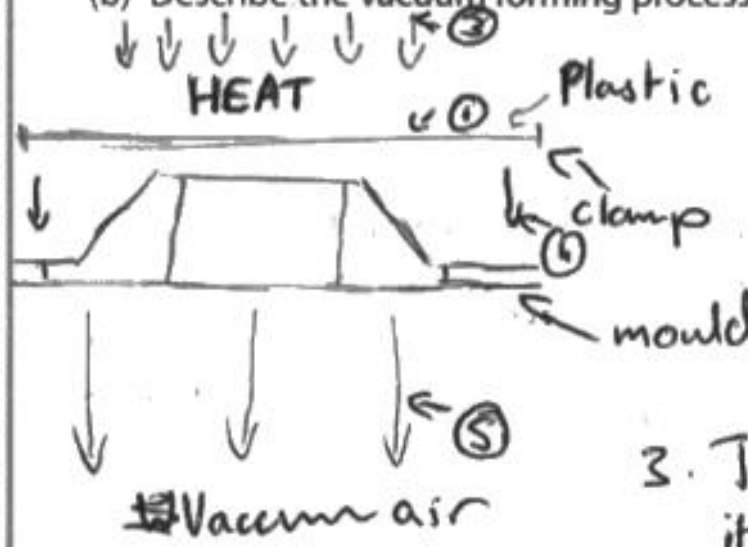
		inappropriate throughout.
Level 2	3 – 4	• Drawing is produced with some precision and accuracy. • Mostly appropriate use of isometric grid with mostly accurate orientation • Some attention to scale, size and dimensions • Line style is mostly appropriate and consistent throughout.
Level 3	5 - 6	• Drawing is produced with precision and accuracy with upper corner points correctly plotted and positioned. • Fully appropriate use of isometric grid with correct orientation • Accurately drawn to scale • Line style is appropriate and consistent throughout enhancing the 3D effect.

Note – acceptable to correctly show the top flaps in an open position.


A
B
C

Describe processes

(b) Describe the vacuum forming process, using annotated sketches.



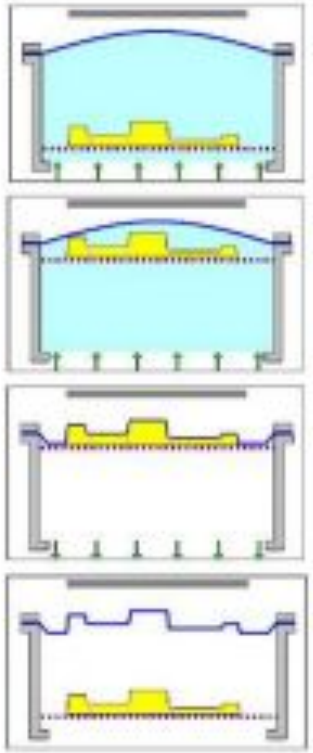
Q03b 4

1. plastic sheet is cut to size and clamped in the vacuum former
2. ~~painting~~ release agent is applied to the mould
3. The plastic sheet is heated until it ~~has~~ sat
4. The plastic sheet is lowered onto the mould
5. The vacuum sucks any resided air between the plastic and mould
6. mould is cooled
7. mould is seperated from the plastic that has been formed
8. Excess material is removed

Question number	Answer	Mark
3(b)	Award marks as follows (maximum 4 marks): 1. The mould is placed on to the bed / table / platen of the vacuum forming machine (1) 2. Clamp the vacuum forming (polymer/plastic) sheet in position (1) 3. The plastic is heated / the heater is pulled over the plastic until the plastic softens and sags (1) 4. Air is blown in (1) 5. The mould / bed / table is raised / plastic lowered (1) 6. Air is removed using the vacuum pump / the plastic is sucked around the mould (1) 7. Mould is removed / the plastic has cooled (1)	(4)

Process

1 Heat up plastic (HIPS)



2 Remove heater and blow air in

3 Raise the Mould

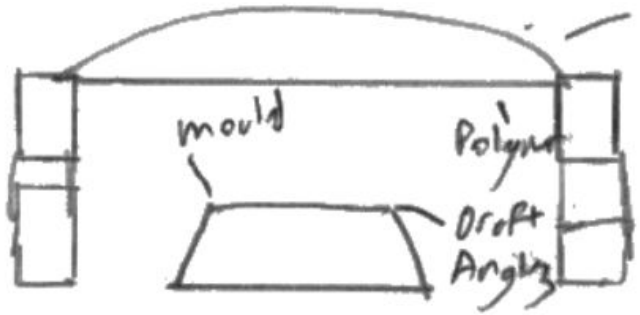
4 Remove air using vacuum pump

5 Lower the mould releasing the shaped plastic

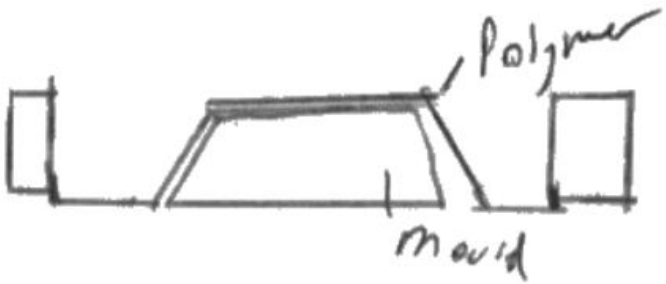
If no sketch, a sketch without annotations or incorrect order, award a maximum of 3 marks.

(b) Describe the vacuum forming process, using annotated sketches.

(4) Q03b 2



Heater heats Polymer until Pliable



Polymer brought down onto the mould and Air Sucked out to Prevent warping.

(b) Describe, using labelled sketches, how a square section of mahogany would be prepared and then turned into a cylinder, using a wood lathe.

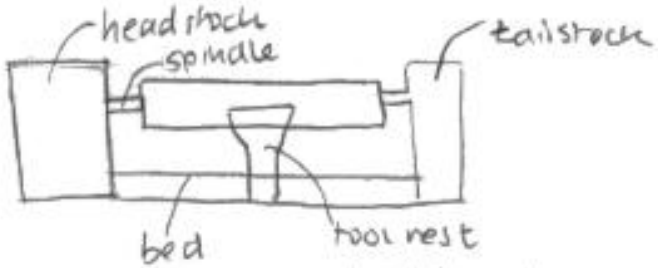
(4) Q03b

① The mahogany would be cut into a cuboid with a square cross section.

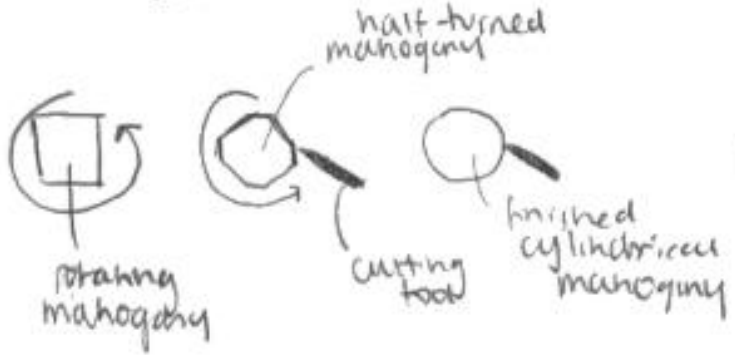


② The mahogany is fixed into the lathe and the lathe is turned on.

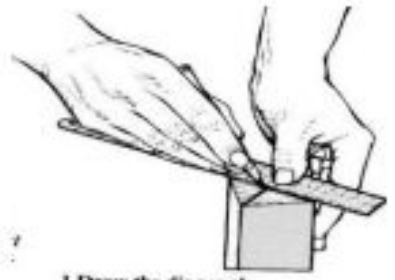
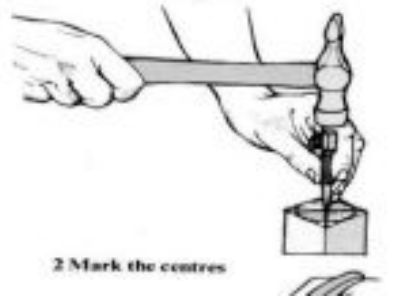
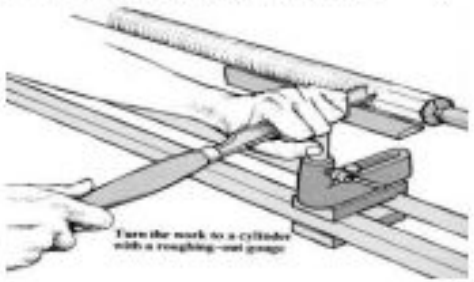
③ The motor in the headstock rotates the spindle and the mahogany turns.



④ Guided by the tool rest, a cutting tool is pushed into the mahogany, removing layers until the desired profile is achieved, in this case a cylindrical shape



- A labelled diagram of the machine/process is useful but must also have a description of the process to go with the sketch.
- No marks for labelling all the parts.

Question number	Answer	Mark
3(b)	Award marks as follows (maximum 4 marks): 1. Marking centre / diagonals (1) 2. Cutting along the diagonals to embed drive centre / punch centre (1) 3. Plane to an octagonal section / to remove corners (1) 4. Mounting the workpiece in the lathe / between centres (1) 5. Positioning the tool rest (1) 6. Workpiece/mahogany rotates (1) 7. Initial cut using roughing-out gouge/insert (1) 8. Smooth with a skew chisel/insert (1) 9. Sanding (1) 10. Parting-off (1)  1 Draw the diagonals  4 Plane the stock to an octagon  2 Mark the centres 	(4)

Maths Questions

(b) The graphics on the packing case will be printed using black ink.

Each packing case requires 2.5 ml of ink.

Ink is supplied in 1.5 litre containers.

The manufacturer needs to print graphics on 3,500 packing cases.

The manufacturer buys sufficient full containers of ink to complete the print run.

Calculate how many extra packing cases the manufacturer could print before running out of ink.

(5)

Question number	Answer	Additional Guidance	Mark
2(b)	In order for the candidates to solve the problem, they will need to recognise that each of the following stages are required. <u>Method 1</u> Stage 1 (calculate the total volume of ink required) 3500 x 2.5 = 8750 ml **(1)** Stage 2 (convert to litres) 8750/1000 = 8.75 litres **(1)** Stage 3 (calculate number of containers of ink required) 8.75/1.5 = 5.8333… rounded to **6 full containers (1)** Stage 4 (calculate ink remaining) (6 x 1.5) – 8.75 = 0.25 litre **(1)** Stage 5 (calculate number of extra printed cases) (0.25 x 1000/2.5 = 100 **(1)**	Accept alternative methods of correct working out. Error carried forward should be applied. Award full marks for correct answer only. For stage 3 to be awarded the answer must be 6 Conversion of units may be applied at any point.	**(5)**

Students must show all their working-outs as there are marks available via ECF if they have used the correct method/stages

Method 2

Stage 1 (convert l to ml)
1.5 x 1000 = 1500 **(1)**

Stage 2 (calculate the number of prints from one container)
1500/2.5 = 600 **(1)**

Stage 3 (calculate number of containers requited)
3500/600 = 5.83 = 6 full containers **(1)**

Stage 4 (calculate max number of prints from 6 containers)
6 x 600 = 3600 **(1)**

Stage 5 (calculate the number of extra prints)
3600 – 3500 = 100 **(1)**

(5 Q02b 5

remember

2.5×3500 = 8750 ml = 8.750 Litres of ink

$\frac{8.750}{1.5} = 5.8\dot{3}$ containers needed = $\frac{35}{6}$

∴ 6 containers ordered

remainder of ink $6 - \frac{35}{6} = \frac{1}{6}$

$\frac{1}{6} \times 1.5 = 0.25$ litres remaining = 250 ml

$\frac{250}{2.5} = 100$ remaining cases

Answer 100 cases

(Total for Question 2 = 7 marks) Total 7

(b) The graphics on the packing case will be printed using black ink.

Each packing case requires 2.5 ml of ink.

Ink is supplied in 1.5 litre containers.

The manufacturer needs to print graphics on 3,500 packing cases.

The manufacturer buys sufficient full containers of ink to complete the print run.

Calculate how many extra packing cases the manufacturer could print before running out of ink.

(5) Q02b 3

2.5ml = 2.5ml (= 0.01L ÷ 4 × 100 = 1L)



× 3,500

 = 1.5L

100 × 1000

2.5 × 3,500 = 8750ml = 87.5 Litres

÷ 1000 8.75 litres

5 = 7.5 59

6 = 9L ÷100

59.

Answer 59 & 6 cases

Calculate how many extra packing cases the manufacturer could print before running out of ink.

(5) Q02b 3

2.5ml = 2.5ml (= 0.01L ÷ 4 × 100 = 1L)



× 3,500

2.5ml

 = 1.5L

100 × 1000

2.5 × 3,500 = 8750ml = 87.5 Litres

÷ 1000 8.75 litres

5 = 7.5 59

6 = 9L ÷100

59.

5 A contractor has been asked to paint the floor of a design and technology workshop.

Figure 4 shows the floor plan of the design and technology workshop.

All dimensions are in metres (m)

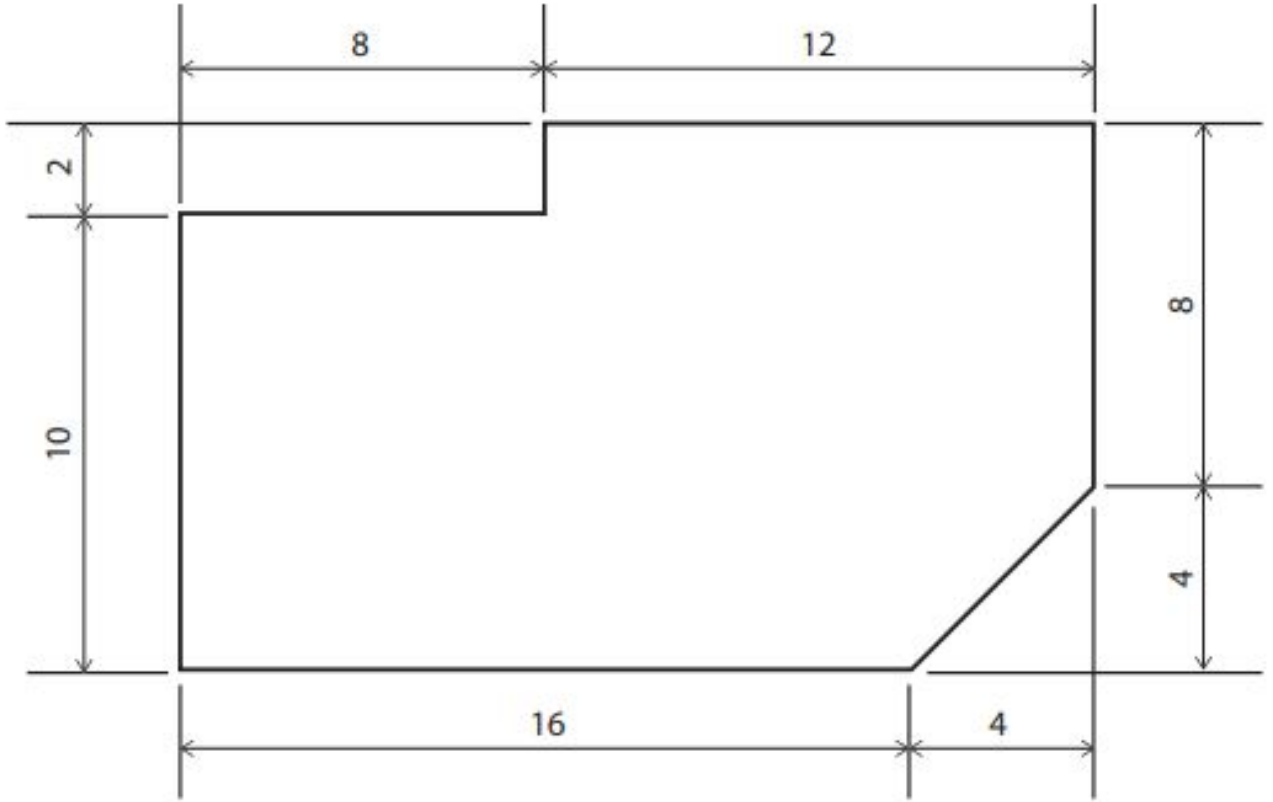


Figure 4

(a) Calculate the area of the design and technology workshop floor.

Give your answer to **one decimal place**.

(5)

5 A contractor has been asked to paint the floor of a design and technology workshop.

Figure 4 shows the floor plan of the design and technology workshop.

All dimensions are in metres (m)

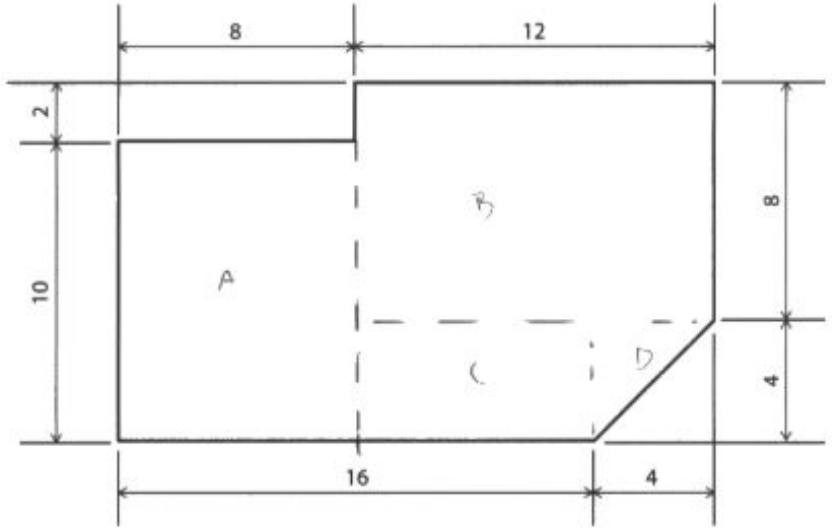


Figure 4

(a) Calculate the area of the design and technology workshop floor.

Give your answer to **one decimal place**.

(5) Q05a 5

A = 8 × 10 = 80 m²

B = 8 × 12 = 96 m²

C = 8 × 4 = 32 m²

D = $\frac{4 \times 4}{2}$ = 8 m²

80 + 96 + 32 + 8 = 216 m²

Answer 216.0 m²

Question number	Answer	Additional Guidance	Mark
5(a)	Method 1 Stage 1 (calculate area of full rectangle) **(1)** $(8+12) \times (8+4) = 240m^2$ Stage 2 (calculate area of rectangular cut out) **(1)** $8 \times 2 = 16m^2$ Stage 3 (calculate area of triangular cut out) **(1)** $4 \times 4 / 2 = 8\ m^2$ Stage 4 (calculate area of floor) **(1)** $240 - 16 - 8 = 216m^2$ Stage 5 (shown to 1dp) **(1)** Area = $216.0m^2$	Accept alternative methods of correct working out. Error carried forward should be applied. Award full marks for correct answer only (to 1DP). SC 216 seen award 4 marks	**(5)**

Method 2

Stages 1/2
(rectangle 1)
$8 \times 10 = 80m^2$

(rectangle 2)
$8 \times 12 = 96m^2$

(rectangle 3)
$4 \times 8 = 32\ m^2$

2 marks for all 3 correct rectangles
1 mark for 2 correct rectangles

Stage 3
(triangle) **(1)**
$4 \times 4/2 = 8m^2$

Stage 4 **(1)**
$80 + 96 + 32 + 8 = 216m^2$

Stage 5 (shown to 1dp) **(1)**
Area = $216.0m^2$

(b) The floor paint is only supplied in 5 litre cans.

A 5 litre can costs £48.00.

One litre of paint will cover 12 m^2 for each coat.

Two coats are required.

Calculate the overall cost of the cans of paint required to complete the painting of the floor.

(3)

Question number	Answer	Additional Guidance	Mark
5(b)	<u>Method 1</u> Stage 1 (calculate paint required to complete two coats) **(1)** (216/12) x 2 = 36 litres Stage 2 (calculate number of cans of paint required) **(1)** 36/5 = 7.2 = 8 cans Stage 3 (cost of paint) **(1)** 8 x 48 = £384.00 (accept £384) <u>Method 2</u> Stage 1 (calculate coverage of 1 can) 12 x 5 = 60m^2 Stage 2 (calculate number of cans for 2 coats) (216 x 2)/60 = 7.2 = 8 cans Stage 3 (cost of paint) **(1)** 8 x 48 = £384.00 (accept £384) (Note work with candidates own floor area under ecf as a substitution for 216)	Accept alternative methods of correct working out. Error carried forward should be applied (including from part (a) ie work with the candidates own figure for area). Award full marks for correct answer of £384 or using their own starting point for the floor area.	**(3)**

(b) The floor paint is only supplied in 5 litre cans.

A 5 litre can costs £48.00.

One litre of paint will cover 12 m² for each coat. ✓

Two coats are required.

Calculate the overall cost of the cans of paint required to complete the painting of the floor.

(3) Q05b 3

216 m.² ÷ 12 = 18

need 18 litres

18 x 2 = 36 litres

need 8 x (5 litres)

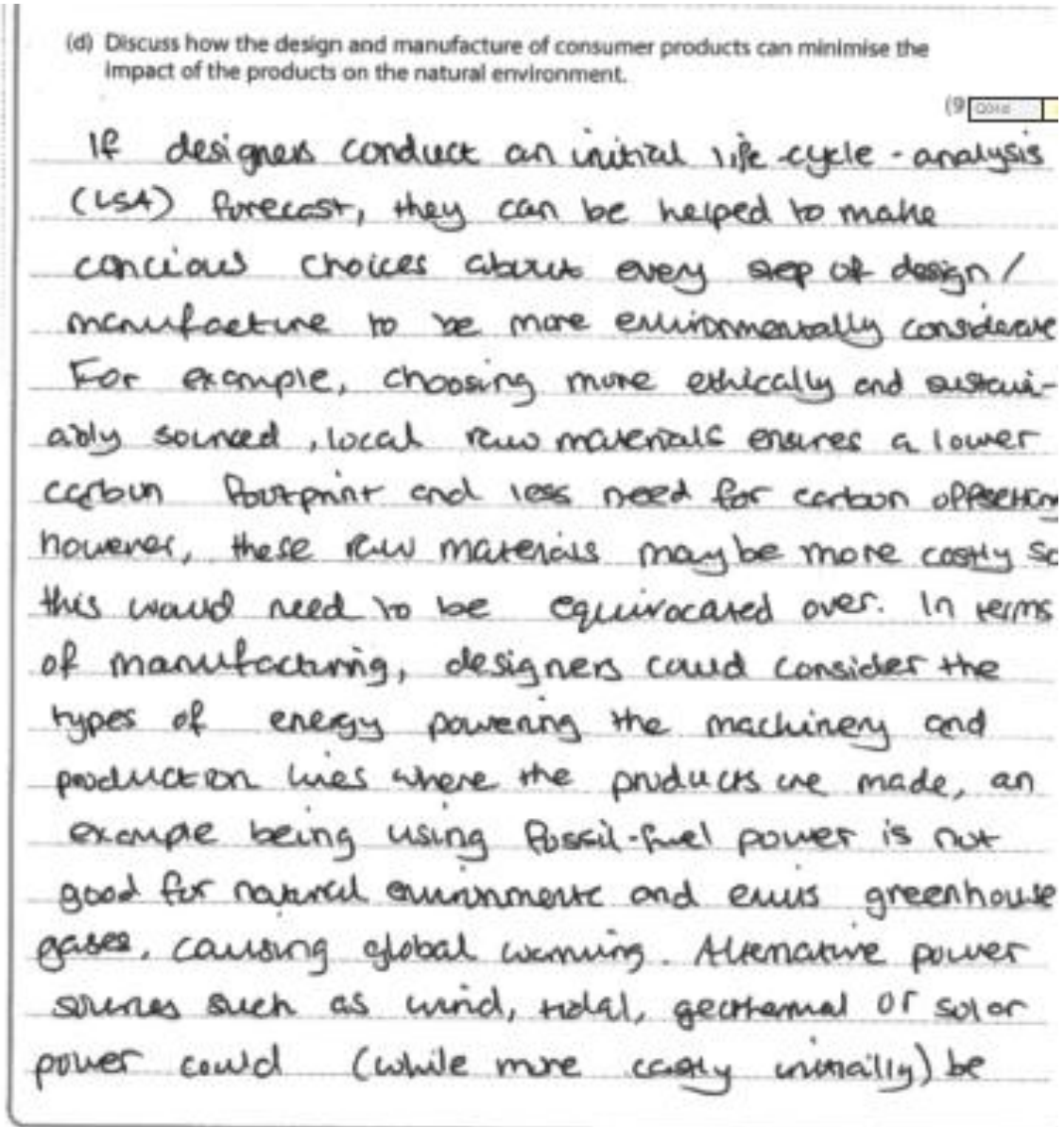
8 x 48.00 = £384

Answer £ 384

(Total for Question 5 = 8 marks) Total 8

- Students must show all their working-outs as there are marks available via ECF if they have used the correct method/stages.
- If the student had calculated part a incorrectly and then brought that answer into the start of their part b answer it was still possible to achieve 3 marks if they used the correct method.

Long answer/Discuss/Outline Questions

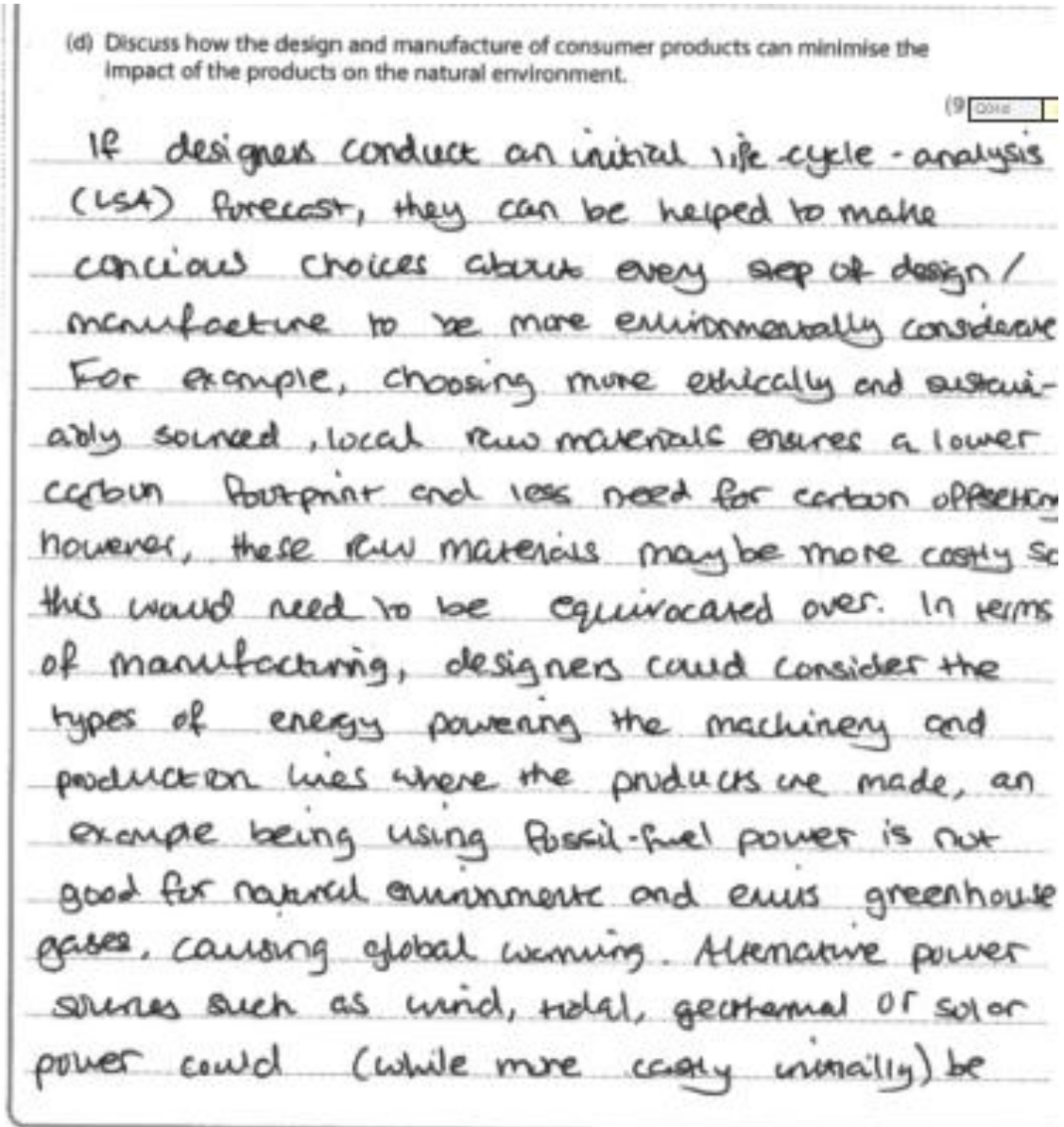


(d) Discuss how the design and manufacture of consumer products can minimise the impact of the products on the natural environment.

(9)

If designers conduct an initial life cycle - analysis (LCA) forecast, they can be helped to make conscious choices about every step of design / manufacture to be more environmentally considerate For example, choosing more ethically and sustainably sourced, local raw materials ensures a lower carbon footprint and less need for carbon offsetting however, these raw materials may be more costly so this would need to be equivocated over. In terms of manufacturing, designers could consider the types of energy powering the machinery and production lines where the products are made, an example being using fossil-fuel power is not good for natural environments and emits greenhouse gases, causing global warming. Alternative power sources such as wind, tidal, geothermal or solar power could (while more costly initially) be

Long answer/Discuss/Outline Questions



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much more environmentally friendly. Hence, designers can design the plants and packaging machinery to be powered sustainably. The packaging materials should also be considered and more ethical, renewable ones should be chosen, this may however increase unit cost of the product. The distribution channels chosen and transport modes must be considered by designers, the size, weight and market of the product would mean they could design with an ethical method of logistics in mind. For example, using delivery via electric vehicles which don't directly contribute to global warming. Also in the LSA is use, for example standard components can be included in the design to be easily replaced / fixed by consumer to avoid premature disposal. In terms of disposal, recyclable materials should have been used so they can be separated at a recycling plant and not simply disposed of in waste and landfill, causing chemical pollution and hence being environmentally detrimental.

(Total for Question 4 = 20 marks)

Total

- Students must produce detailed, thorough answers that discuss many of the relevant points.
- Relevant points should be explained so the use of bullet points is discouraged unless each point is detailed.

Levels-based mark schemes

Question number	Answer	Mark
4(d)	This question asks candidates to discuss how the design and manufacturing of consumer products can minimise the impact of the product on the natural environment. Candidates might refer to the following in their responses: Material selection • source • quantity • quality • range • recyclability • biodegradability Manufacture • minimising energy use • optimum use of materials / components • simplification / streamlining of processes • minimising waste • scale of production Distribution • efficient / minimisation of packaging • maximisation of product carried on each vehicle Use • energy efficient products • ease of repair / maintenance End of life • design for disassembly • recovered material feasibility • potential for re-processing • potential for energy recovery • minimising elements to landfill	(9)



Level	Mark	Descriptor
	0	No rewardable materials
Level 1	1 – 3	• Superficial discussion that considers a narrow range of factors, demonstrating limited understanding. • Partial application of understanding to the context of the question.
Level 2	4 – 6	• Coherent discussion that makes some relevant links between a sufficient range of factors, demonstrating competent understanding. • Generally sound application of understanding to the context of the question.
Level 3	7 - 9	• Comprehensive discussion that makes effective links between a wide range of factors, demonstrating thorough understanding. • Considered and effective application of understanding to the context of the question.

(d) Discuss how the design and manufacture of consumer products can minimise the impact of the products on the natural environment.

(9) Q04d 5

sourcing materials, recyclability, longevity, energy efficiency

When manufacturing consumer products, designers should be locally sourced and biodegrade to reduce the carbon footprint of the product. This minimises the impact of the product on the natural environment, as less carbon was used to manufacture the product.

The designer should consider the longevity of the product, making parts replaceable with no designed obsolescence. These factors will reduce products thrown away, possibly tipped into and damaging the natural environment.

The end of life cycle part of a products life should be considered by the designer, ensuring the product can be recycled, repurposed, reused or biodegradable easily will ensure less of an impact on the natural environment.

Manufacturers should take special care when producing possibly more consumer products, focusing on energy efficiency, reducing energy wastage wherever possible. This will reduce the impact of the commercial product on the natural environment as less fuel would have to be consumed in production.

Final Question / Product analysis / Evaluate / 12 marks

11 Figure 8 shows a bedside lamp.



Figure 8

- The lamp can be switched on or off by touching either the base or tubular support.
- Three levels of lighting are available, by touching the base or support one, two, or three times.
- The lamp switches off after the fourth touch of the base or support.
- The lamp can use either LEDs or halogen bulbs.
- The base is made out of stainless steel and has rubber on the underside.
- The lamp weighs 2.7 kg.
- The overall height of the lamp is 330 mm.
- The diameter of the lamp base and shade is 130 mm.

Evaluate the functionality of the bedside lamp with reference to aesthetics and user requirements within a home setting.

(12)

Level	Mark	Descriptor
	0	No rewardable materials
Level 1	1 – 3	• Applies a basic understanding to deconstruct information, making limited connections between concepts. • Incomplete evaluation with unresolved conclusion that demonstrates limited synthesis of understanding. • Judgements are tentatively supported by evidence.
Level 2	4 – 6	• Applies a generally sound understanding to deconstruct information and provide some clear connections between concepts. • Imbalanced evaluation that synthesises some relevant understanding into a generally coherent conclusion. • Judgements are occasionally supported by relevant evidence.
Level 3	7 - 9	• Applies an effective understanding to deconstruct information and provide logical connections between concepts. • Balanced evaluation that synthesises relevant understanding into a considered conclusion. • Judgements are mostly supported by relevant evidence.
Level 4	10 - 12	• Applies a comprehensive understanding to deconstruct information and provides insightful connections between concepts throughout. • Thorough and balanced evaluation that synthesises relevant understanding into a well-developed conclusion. • Judgements are supported by pertinent evidence throughout.



(12)

The open top is great for functionality as it not only allows the light to travel further into the home (aiding user convenience) but aesthetically it also creates a modern, sleek, open look which is visually appealing. The exposed bulb may however not be aesthetically ideal for some users who may perceive it to be ugly. The frosted outer surface allows a more mellow, ambient transmission of light into the room in the home setting, yet the open top lets light flow above, this depends however on where the lamp is positioned within the home and the height of the user in relation to it. The 15mm thick glass on the lamp gives the user an impression of quality due to its 2.7kg weight and luxurious visual elements, perfect for an stylistic home setting to be aesthetically attractive for the users to feel comfortable at home. The curved, sleek steel support is aesthetically pleasing for the user, yet also functional with the ability to switch it on or off by touching. This is good as the switch is very large and the user may be in bed so sleepy and doesn't need to find a precise switch location to switch it on or off. Also, the base is made from stainless steel, which is polished to be aesthetically pleasing yet may easily be scratched, especially when routinely touched (being the switch) by the user. This may also add to the weight and could possibly make the lamp too heavy for easy transportation, retracting from user convenience therefore. The mains cable supply and 3 pin plug socket means that most UK homes could have myriad outlets for the lamp, aiding user convenience and functionality within the home environment. Mains also mean the lamp has passed (presumably) BSI inspection, and is hence safe to use for the user. The varying levels of light mean that the user has multiple modes to suit their light opacity needs (eg. low for night reading, high for general use), meeting their requirements successfully hence. However, touching the lamp by mistake when near or cleaning around it could be tedious for the user as it would switch on and must be pressed another 3 times to turn it off, this could be irritating. The option to use LED's is great for environmentally-conscious or budget-conscious consumers as it's more energy efficient, yet being able to also use the traditional halogens is still convenient. Leaving the lamp on standby constantly may use lots of energy without even turning it on, but using the wall switch may be tedious to users, retracting functionality hence. The rubber on the base is great as it will help non-slip features in case knocked by the user, and if it fell it may cause serious damage due to the weight. In essence, through visual and many functional aspects, the lamp is perfect for a home setting and meets a lot of user requirements, however, some elements such as energy usage and weight may be non-beneficial to the user and cause them to dislike the lamp as a result.

(Total for Question 11 = 12 marks)

TOTAL FOR PAPER = 120 MARKS

Evaluate the functionality of the bedside lamp with reference to aesthetics and user requirements within a home setting.

(12 marks)

The lamp has been designed in accordance with Bauhaus design. The emphasis lies on the functionality over form and the use of tubular steel. The Bauhaus element of the lamp allows for the functionality to be the main feature.

For example, the touch ability to allow for different lighting modes allows for total user control - whether the user may want the room to be dim or bright.

The touch anywhere aspect also allows for total user control as the user does not have to look for a switch to turn the lamp on or off. They can simply turn it on or off with a touch.

The functionality is also shown through the build as the use of stainless steel allows for the lamp to be scratched without leaving a mark.

Also the rubber on the underside acts as a protection for the bedside table or desk.

The metal cannot scratch it.

The open top nature allows for the room to be more brightly illuminated and the frosted glass allows for no shadows to be cast like a normal lampshade.

The heavy 2.7 kg weight allows for the lamp to be sturdy in its place and for it to not move around too much. This is especially good as to control the lamp you need to touch it.

The versatility of either LED bulbs or halogen bulbs allow the user to not be constrained using the lamp.

The curved stem may allow for an easier reach for the user depending on positioning.

11 Figure 8 shows a 'spinning' bike used for indoor group exercise classes.



- Main frame in welded mild steel with a powder coated finish
- Four height adjustable feet at ground contact points
- Seat adjustable vertically and horizontally
- Handlebars adjustable for height
- Digital liquid crystal display showing revolutions per minute (RPM), power output in Watts, distance travelled, heart rate and effort level
- Tension/effort level adjustment
- Stainless steel flywheel – weight 22 kg
- Overall weight – 64 kg
- User height range 1.50 m to 1.95 m
- Maximum user weight – 135 kg
- Cost £1,150.00

- Students must produce detailed, thorough answers that discuss many of the relevant points.
- Identify the key points within the question.

Evaluate how the designer of the 'spinning' bike has addressed ergonomics and user requirements for group exercise activities.

(12)

The spinning bike has an welded mild steel frame, this makes it very durable and tough meaning it will be able to withstand lots of use. The powder coated finish is much more durable than a paint finish therefore metal will be protected for longer especially when things like sweat could enhance the wear of the metal finish. The ability to adjust all 4 gear means the bike will be stable and the user is able to have a comfortable ride wherever they are. This links to ergonomics as it ensures it is comfortable to use on a variety of surfaces. The adjustable seat and handlebars, also mean the bike is accessible for many people and users can customise the bike to suit their own ergonomic requirements. Adjustable components are probably the biggest factor in making something comfortable to use as every person is different and likes their seat angle, height etc. differently. So customisation is a very good thing to have. The LCD display is very informative and helps immerse the rider in the experience and would definitely improve customer satisfaction. Riders are able to view many different metrics, and this is what a person who is interested in fitness would be looking for. The tension / effort level is a very good feature as it means people with a wide varying level of strength / fitness can use the bike. The designer has aimed it to suit as many people as possible, and this is especially important in group exercise as you want everyone to join in and have a sufficient challenge.

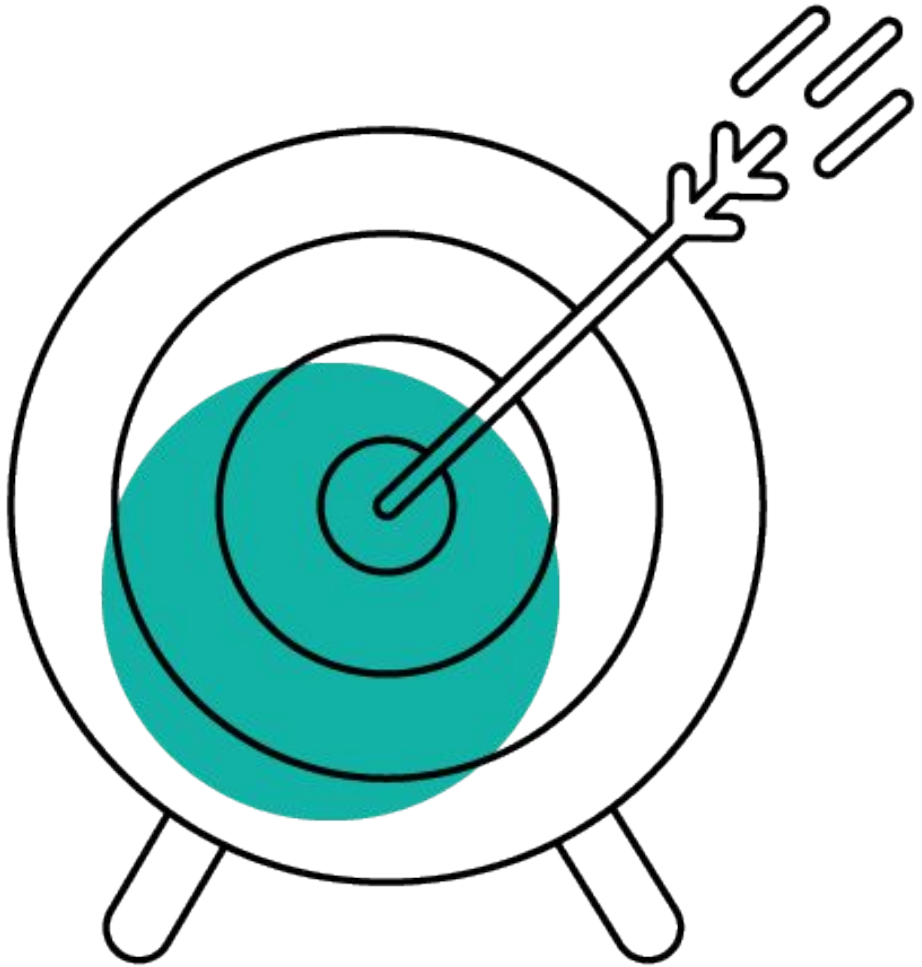
The overall weight of the bike could be seen as very heavy but is easily transported by 2 people. A heavy bike will ensure a stable and safe ride, ~~allowing people~~ Especially in group exercise, it is important to ensure everyone feels comfortable during exercise and the ergonomics of a product are very important. The designer has done very well to ensure their bike is able to be used by as many people as possible at varying height, weight, fitness level etc. The bike is pretty expensive and especially if buying for a group of people, it could easily reach over £10,000 for the bikes. However they are receiving a high quality, well finished bike, which is easy, informative and comfortable to use for all people. The designer has addressed ergonomics by ensuring that the bike can be easily adjusted to suit different people. Overall, the designer has done a very good job of designing the bike. ~~It~~ They have thought about many different aspects of how someone might be different to another and made them parts adjustable, however, there are still some limits on the weight restriction, which could prevent very overweight people from trying to get fit.

(Total for Question 11 = 12 marks) Total 12

Summary

In this session we have covered:

- how to approach the various different types of question on the paper
- tips on how students should answer each type of question and how they could prepare
- ensuring all the specification topics are covered as many of the questions are focused and factual.



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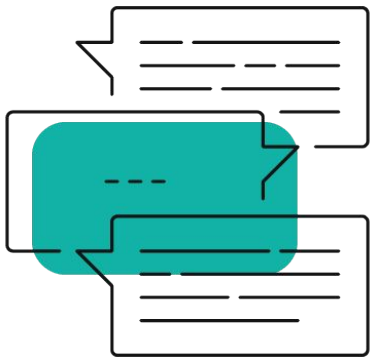
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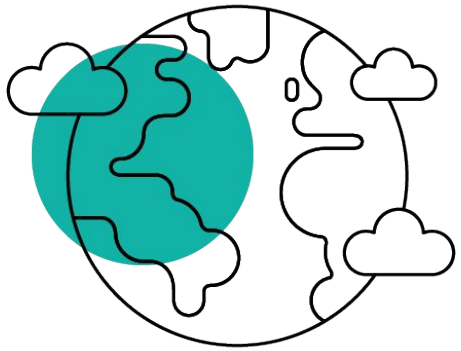


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